

Using a dynamic energy budget model to investigate the physiological mode of action of lead (Pb) to *Lymnaea stagnalis*

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ARTICLE INFO

Keywords:

Lead (Pb)

Lymnaea stagnalis

Dynamic energy budget

DEBtox

Ecological modelling

ABSTRACT

Lymnaea stagnalis is a notably sensitive species for a variety of metals, including lead (Pb). However, the mechanism(s) of lead toxicity to *L. stagnalis* currently remain incompletely understood. Under dynamic energy budget (DEB) theory, different physiological modes of action (PMoAs) result in the emergence of distinct changes to the life histories of exposed organisms. This work aims to better understand the PMoA of lead toxicity to *L. stagnalis* by applying DEB modeling to previously published datasets. After calibration, the model was utilized to evaluate the relative likelihood of several PMoAs. Assuming decreased assimilation, the *L. stagnalis* DEB model was able to capture most, but not all, trends in experimentally observed endpoints, including growth, reproduction, and food ingestion. The weight-of-evidence suggests that decreased assimilation via a decrease in food ingestion is the most plausible PMoA for chronic lead toxicity in *L. stagnalis*. Collectively, our results illustrate how mechanistic modeling can create added value for conventional individual-level toxicity test data by enabling inferences about potential physiological mechanisms of toxicity.

1. Introduction

Lead is released into aquatic environments through a variety of mechanisms, both anthropogenic and natural. Sources of lead in the environment include mining and smelting activities, lead from activities such as hunting and fishing, and decades of use in gasoline (World Health Organization 1995; Pattee and Pain 2003). While lead is a non-essential element, it is crucial in the manufacturing of a wide range of products such as batteries (May et al., 2018). In aquatic environments, lead is a known toxicant to both fish and aquatic invertebrates.

Lymnaea stagnalis is an especially sensitive species for a variety of metals, with particularly high sensitivity to lead having been observed (Amorim et al., 2019). *Lymnaea* snails in general are among the most sensitive investigated freshwater organisms to lead exposure (Borgmann et al., 1978; Grosell et al., 2006; Grosell and Brix 2009). Negative effects on the growth of *L. stagnalis* have been observed with exposure to lead concentrations as low as 1.0 µg/L (Munley et al. 2013). In addition to concern surrounding the sensitivity of *L. stagnalis* to lead, the exact mechanism of toxicity remains unclear (Brix et al., 2012). As such, an expanded understanding of the sensitivity of *L. stagnalis* is important in establishing environmentally safe concentrations of lead.

In fish, lead acts as a Ca²⁺ antagonist and has also been observed to inhibit the uptake of other essential elements, such as Na⁺ and Cl⁻. Lead toxicity also manifests in fish through developmental abnormalities and neurotoxicological effects (Brix et al., 2012). While numerous studies have investigated the effects of lead toxicity in freshwater fish, the mechanisms of lead toxicity in aquatic invertebrates are notably less well studied (Brix et al., 2012; Grosell et al., 2006; Mager 2012). At an ionic level, disruption of Na⁺ homeostasis has been noted in juvenile *L. stagnalis* exposed to lead, although the pattern of this relationship appeared to be complicated (Grosell et al., 2006). Lead-induced inhibition of Ca²⁺ uptake has also been hypothesized to be a process leading to growth inhibition in freshly hatched *L. stagnalis*; however, additional work has suggested that this inhibition may only be a secondary effect (Grosell and Brix 2009; Brix et al., 2012). At a higher level of biological organization, Brix et al. (2012) also observed a decrease in the feeding rate of *L. stagnalis* with exposure to lead, suggesting that decreased ingestion of food may contribute to toxicity in the snails. However, this observation is complicated by a decrease in the growth rate of snails at lead concentrations below those which impacted snail feeding.

Under dynamic energy budget (DEB) theory, different physiological modes of action (PMoAs) result in distinct changes to the life histories of

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exposed organisms (Martin et al., 2012; Jager et al., 2013; Jager, 2017; Ashauer and Jager 2018). DEB models are mechanistic models and provide a mathematical framework connecting chemical stress at the physiological level to organism level effects. Consequently, DEB models can utilize existing data to identify the most likely physiological modes of action (PMoA) for a substance to a given species (Vlaeminck et al., 2019). This work aims to better understand the PMoA of lead toxicity to *L. stagnalis* by applying DEB modeling to previously published datasets.

2. Materials and methods

2.1. Data for model calibration and validation

In order to examine the impacts of lead exposure on the life history of *L. stagnalis*, we centered our work around two focal studies: Munley et al. (2013) and Brix et al. (2012). We chose these studies because both consider multiple endpoints (which can be related to the DEB framework) at multiple time points during exposure. Snails from both studies originated from the same culture, and both experiments were maintained in dechlorinated City of Miami water. In both studies, snail were fed a mixture of lettuce and sweet potatoes. However, in Brix et al. (2012) carrots were also incorporated into the snails' diet. Munley et al. (2013) reported lead effects on both snail weight and egg production in a full life-cycle study, starting with freshly hatched snails. Raw data were received from the original authors. Brix et al. (2012) reported effects of lead exposure on both the weight and food ingestion rate of juvenile snails. Raw data values for Brix et al. (2012) were taken from the figures included in the published study using Origin Version 2018 B 64-bit software (OriginLab, USA).

2.2. Model structure

Dynamic energy budget (DEB) theory uses a set of ordinary differential equations to describe how energy, acquired by an organism through food consumption, is allocated to a variety of life-history processes (Kooijman 2010). These processes include maintenance, growth, maturation, and reproduction within an individual. Energy is treated as a finite resource within DEB models, respecting the law of conservation of energy (Kooijman 2010; Ashauer and Jager 2018). Generally speaking, differences in the allocation of energy between species are described by a set of species-specific DEB parameters (Table 1). These parameters are used to calculate the dynamics of the three DEB state variables: structural length, reserve, and maturity, using standard DEB equations. These equations are implemented in the present study as they are presented in Zimmer et al. (2014). DEB principles can be applied to describe energy allocation within any organism, with some particular alterations to describe major discrepancies in life histories. For example, in developing a full DEB model for *L. stagnalis*, Zimmer et al. (2013, 2014) noted that the growth curve of *L. stagnalis* differs from the von

Bertalanffy (1934) growth curve and incorporated a submodel to account for metabolic acceleration in juvenile *L. stagnalis*. This metabolic acceleration submodel was also incorporated in the analysis presented here through the use of the 'abj' DEB model.

2.3. Calibration to control data

Before the effects of lead exposure were considered, the model was tuned to data from control snails in each study. This was done in order to ensure the model adequately captured trends in the growth and reproduction of each of the snails in their specific experimental conditions in absence of lead. Snails originated from the same culture in both studies, therefore calibration to control data should primarily account for differences in experimental design and any potential differences in food quality. For each of the focal studies independently, previously established DEB parameters for *L. stagnalis*, as presented in the Add-my-pet database (Table 1), were used as a starting point. Additionally, both a temperature correction and an assimilation factor were applied. Temperature correction was implemented because both focal studies were conducted at temperatures that differ from the standard DEB temperature (20 °C). The temperature correction was implemented using the Arrhenius temperature (Eq. (1)), as has been done in earlier *L. stagnalis* studies (Zimmer et al., 2014; Freitas et al., 2007). The temperature correction factor is dependent on the reference temperature T_{ref} (293 K) and the Arrhenius temperature T_A (Table 1)

$$k(T) = \exp(T_A / T_{ref} - T_A / T) \quad (1)$$

The assimilation factor is an additional parameter intended to correct for differences in food sources, genetic strain, and/or water physicochemistry between the two focal studies and those used in the calibration of the Add-my-pet DEB parameters for *L. stagnalis*. Similar approaches have been widely applied in DEB-IBMs to account for differences between experimental setups, with this additional parameter being termed the assimilation factor, the assimilation correction factor, or the food factor, depending on context (Pereira et al., 2019; Vlaeminck et al., 2019; Zimmer et al., 2012; Zimmer 2013). The assimilation factor (a_f) influences the assimilation flux (S_A) of an organism.

$$S_A = a_f \times f \times p_{Am} \times L^2 \quad (2)$$

The assimilation factor was calibrated for each experimental setup separately, using the software package DEBtool (Marques et al., 2018) implemented in Matlab 9.11. DEBtool was selected for calibration because it allows for automated estimation of multiple DEB parameters, based on simultaneous fitting to multiple life history endpoints (Marques et al., 2018; Koch and De Schampelaere 2019).

The dry weight was estimated to comprise 15% of wet weight, which is comparable to ratios which have been observed in laboratory settings (Ter Maat et al., 2007). The wet weight (including the shell) is assumed to be directly proportional to dry weight, with the shell not being directly accounted for. However, Zimmer (2013) noted that the shell itself comprises only about 10% of the dry weight of a snail. Reproduction was directly compared between model simulations and observations as the cumulative number of eggs produced. In attempting to calibrate the model to the control data for Munley et al. (2013), a large discrepancy was encountered between the egg production observed in the study and that predicted with the Add-my-Pet DEB parameters for *L. stagnalis* (Table 1, Supplemental Information). Dry weight of the snails is estimated in the model as the sum of structure and reserve. The discrepancy between predicted and observed reproductive endpoints (with snails reproducing more than model predictions) persisted after calibration of the assimilation correction factor, even though snail weight over time was reasonably well predicted (Supplemental Information). This deviation in the timing and amount of observed reproduction from what is anticipated from the standard *L. stagnalis* DEB parameters (while weight was reasonably well predicted) suggested a

Table 1

The DEB-parameters for *L. stagnalis* as calibrated by Zimmer (2013). These parameters are available in the Add-my-pet database (Add-my-pet 2016).

Parameter	Symbol	Value	Unit
Maximum surface-specific assimilation rate	p_{Am}	25.2049	J / (d × cm ²)
Maturity threshold at birth	E_H^b	0.285	J
Maturity threshold at transformation	E_H^t	50.1	J
Maturity threshold at puberty	E_H^p	496.6	J
Volume-specific somatic maintenance rate	p_M	58.57	J / (d × cm ³)
Volume-specific cost for structure	E_G	3900	J / cm ³
Energy conductance	v	0.01394	cm / d
Allocation fraction to soma	κ	0.6876	–
Reproduction efficiency	κ_R	0.475	–
Maturity maintenance rate	k_J	0.002	d ⁻¹
Somatic maintenance rate	k_M	0.015	d ⁻¹
Maximum surface-specific searching rate	F_m	6.5	L / (d × cm ²)
Arrhenius temperature	T_A	8000	K

difference in energy allocation to the [Munley et al. \(2013\)](#) snails' reproductive processes, although what prompted this difference is unclear.

A sensitivity analysis indicated that the maturity threshold at birth (E_H^b) was a critical factor in the calibration, that had a strong impact on reproduction as an endpoint. The lower the value of this parameter for an individual, the longer it will take for it to reach the reproductive state. The reduction of the maturity at birth also reduces the costs per egg. A new assimilation factor, to reflect specific experimental conditions, was calibrated simultaneously with an updated maturity threshold at birth ([Fig. 1, Table 2](#)). It should be noted that no biological rationale for a difference in maturity at birth between studies has been identified, and as such should remain a point of caution. However, this approach was preferred over a recalibration of the entire DEB parameter set, in order to avoid ending up with a set of parameter values which were overly-specific to the snails in [Munley et al. \(2013\)](#). For the estimation of these parameters, the Nelder-Mead method ([Nelder and Mead, 1965](#)) was applied in DEBtool, which optimizes the model by iteratively adjusting parameters to minimize a multiplicative symmetric loss function. Further description of this method in an AmP context can be found in [Marques et al. \(2018\)](#).

To calibrate the model for control data from [Brix et al. \(2012\)](#), no additional sensitivity analysis or adjustment of the maturity threshold at birth was necessary. A satisfactory fit to measured weight data could be obtained using only an assimilation factor ([Fig. 1, Table 2](#)). No reproductive data was available for [Brix et al. \(2012\)](#).

Table 2

The calibrated value for the assimilation factor (a_f) and maturity threshold at birth (E_H^b), as estimated using the Nelder-Mead method in DEBtool for [Munley et al. \(2013\)](#) and [Brix et al. \(2012\)](#). The original species parameter value from the Add-My-Pet database (AmP 2019) is included for comparison. The goodness of fit for each set of parameters was evaluated using two different criteria: the mean relative error (MRE) and the symmetric mean squared error (SMSE).

	a_f [-]	E_H^b [J]	MRE [-]	SMSE [-]
Add-my-pet 2016	1.0	0.285	–	–
Munley et al. (2013)	3.379	0.1719	0.138	0.160
Brix et al. (2012)	5.595	0.285	0.050	0.040

2.4. Modelling exposure and sublethal effects of lead

The uptake and elimination processes of lead were modeled using a one-compartment model for the scaled damage level D_w , which has the same unit as the external concentration C_w [$\mu\text{g/L}$] ([Jager 2020; Jager et al., 2023](#)).

$$\frac{d}{dt}D_w = k_d \times \frac{L_m}{L} \times (C_w - D_w) - D_w \times \frac{3}{L} \frac{d}{dt}L \quad (3)$$

with $L_m = \kappa \times (\{p_{Am}\}/[p_M])$

Where k_d is the dominant rate constant [d^{-1}], L_m [cm] denotes a reference length, which is based on the theoretical maximum length of a snail without the effects of chemical stress and without metabolic acceleration, and L [cm] is the actual structural length. Because our

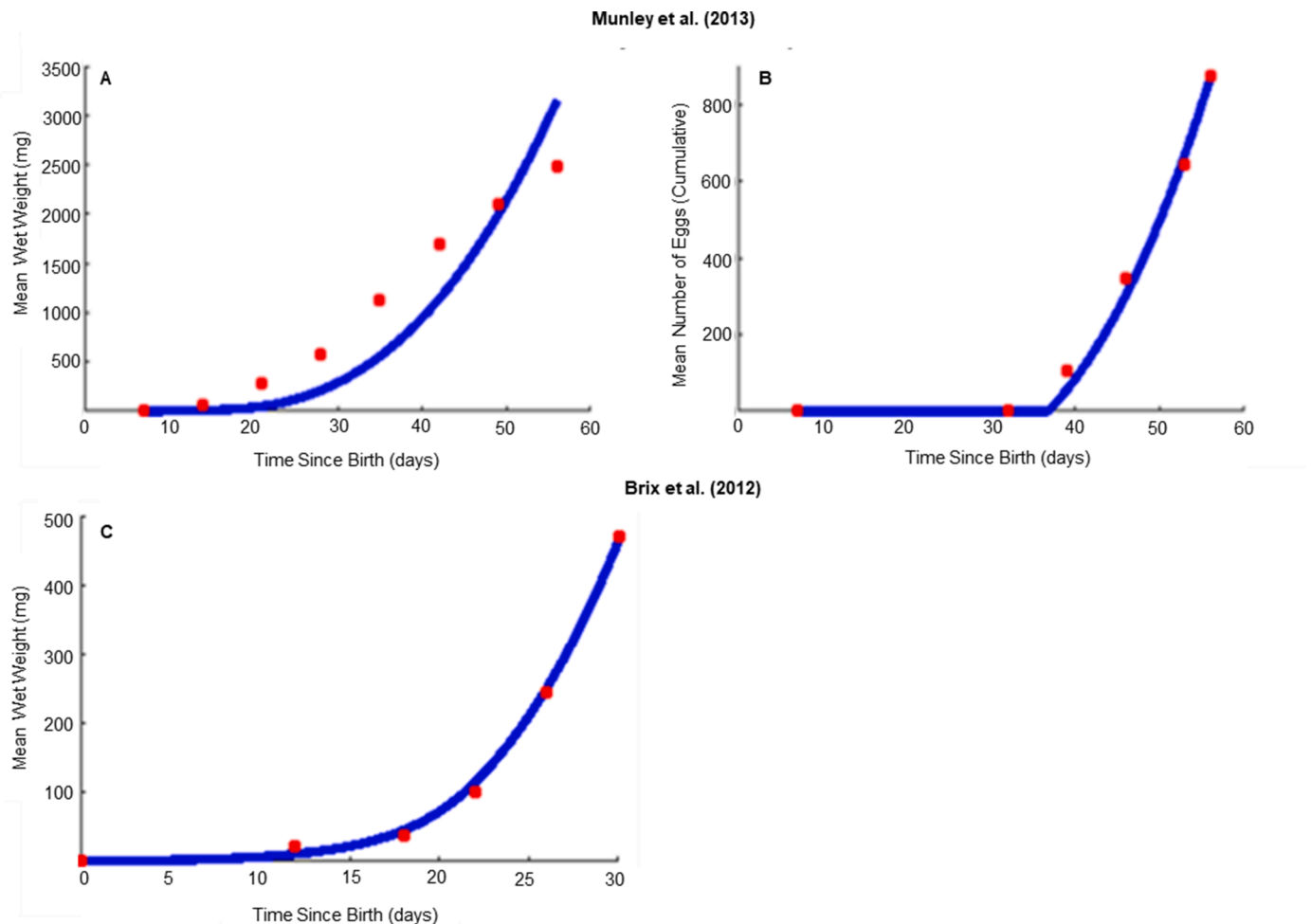


Fig. 1. Observed snail growth (as wet weight) and cumulative egg production in the control treatment (red dots) plotted against individual level model predictions (blue lines) using the appropriate assimilation factors and maturity threshold at birth values for [Munley et al. \(2013\)](#) and [Brix et al. \(2012\)](#), respectively ([Table 2](#)).

toxicokinetic model incorporates only structural length, reserve is not accounted for in our damage dynamics, and structure is assumed to play a more important role in toxicokinetics. This is one possible configuration of the more flexible equation proposed by Jager (2020). The selection of this implementation is largely based on precedent in the DEBtox literature, where such an approach has been standard (Jager and Zimmer 2012). The first portion of the equation accounts for the surface: volume scaling of uptake and elimination of lead, while the latter portion accounts for growth dilution. The generic approach put forth by Jager (2020) also allows for the inclusion of losses through reproduction of eggs, but this has not been accounted for in the present work, as the data used for calibration is comprised of growth and reproduction data. As noted in Jager (2020), such datasets do provide much opportunity to distinguish between different configurations of the scaled damage equation.

The external concentration C_w was assumed to be constant and was set to the mean dissolved lead concentration for each experimental treatment (Jager and Zimmer 2012). It must be noted that uptake and elimination kinetics for lead in *L. stagnalis* are very unlikely to occur via passive diffusion, as in the simple one-compartment model presented here. Crémazy et al. (2019) suggested that the short-term (2–3 h) uptake of lead by *L. stagnalis* may be described using a Michaelis-Menten type relationship, but data regarding elimination are not available. Further examination of how best to describe lead uptake and elimination in *L. stagnalis* is an important area for refinement of the modeling approach described here.

Sublethal effects of lead were implemented following a DEBtox approach using the stress level s [-] (Jager 2020; Martin et al., 2013b). The value of the stress level depends on the scaled damage level, D_w [$\mu\text{g/L}$]. The stress level increases linearly with D_w when the threshold for sublethal effects z_b [$\mu\text{g/L}$] is exceeded. In this function, the slope is defined by the energy budget effect strength b_b [$\text{L}/\mu\text{g}$]:

$$s = b_b \max(0, D_w - z_b) \quad (4)$$

The stress level can be used to modify one or multiple model parameters, and – as a result – the associated life history processes. The life history processes impacted by the stress level are determined by the physiological mode of action (PMoA). We considered three possible PMoAs for lead: (1) a decrease in energy assimilation, (2) an increase in the costs associated with organism maintenance, and (3) an increase in the costs associated with growth. Each of these PMoAs relates the stress value to distinct processes within an organism's energy allocation (i.e. DEB processes) (Table 3).

The standard DEB calls for implementation of maternal effects, in which the cost of an egg and the initial conditions of an embryo depend on the status of the mother. A strict implementation of the maternal effects rule then results in egg costs which change over time and respond to toxic stress. The present study does not strictly adhere to this implementation. The maternal effects rule was used to estimate of the cost of an egg for the imposed control food level per study. Beyond that, maternal effects were ignored and egg costs were assumed to be constant (per study). As such, the cost of an egg was assumed to be independent of chemical stress affecting the mother's reserve. This presents a limitation to our analysis as, if lead toxicity impacts maternal investment per egg,

the model is unable to account for that.

2.5. Calibration procedure and goodness of fit criteria

After assimilation factor calibration, the previously described toxicological part of the model was subsequently calibrated to the sublethal lead toxicity effects observed in Munley et al. (2013) and Brix et al. (2012) independently. The DEBtox model was calibrated based on growth data and reproduction data when available.

The dominant rate constant (k_d), the threshold for sublethal effects (z_b), and effect strength (b_b) were calibrated simultaneously using the Nelder-Mead method. These effect parameters were calibrated for each of the examined PMoAs individually. The likelihood of each PMoA was evaluated using two different goodness of fit criteria as proxies: the mean relative error (MRE) and the symmetric mean squared error (SMSE) (Supplemental Information).

2.6. Validation with effects data not used in calibration

The weight and reproduction of snails over the course of exposure were both utilized in the calibration of DEBtox parameters for lead toxicity. However, additional endpoints such as the age of snails at first reproduction (Munley et al., 2013) and the feeding rate of snails (Brix et al., 2012) were not considered in calibrating toxicity effect parameters. Model predictions for these endpoints were compared to experimental observations in order to further evaluate the validity of each PMoA.

3. Results

3.1. Fits to growth and reproduction data

Of the three examined PMoAs, simulations assuming decreased assimilation provided the best predictions considering both growth and reproductive endpoints from Munley et al. (2013) together (Fig. 2A, Fig. 2D). Decreased assimilation overpredicted observed effects on growth at the highest tested lead concentration (8.4 $\mu\text{g/L}$), but was the only PMoA which captured reproduction effects well. This was reflected in decreased assimilation simulations resulting in both the lowest mean relative error (MRE) and symmetric mean squared error (SMSE) for the Munley et al., 2013 study. It should be noted that simulations assuming increased growth cost produced the lowest error when only growth endpoints were considered (Fig. 2B). However, interpretation of this observation is complicated by the imperfect fit of the model to control growth patterns (Fig. 1). Additionally, increased growth cost was unable to capture the observed impacts of lead on reproduction, where effects on reproduction were underpredicted (Fig. 2E).

Of the three examined PMoAs, simulations assuming decreased assimilation as the PMoA provided predictions closest to observed growth in Brix et al. (2012) (Fig. 3A). While reasonable predictions for snail wet weight could be reached using all three PMoAs, simulations assuming decreased assimilation produced both the lowest MRE and the lowest SMSE.

3.2. Prediction of food consumption effects

In addition to the effects of lead exposure on snail growth, Brix et al. (2012) noted an effect on snail feeding. After 8 and 16 days of exposure, exposed snails exhibited feeding inhibition. The amount of food consumed by exposed snails is reflected in simulations as the feeding flux (S_X), which is directly proportional to the assimilation flux (S_A).

Brix et al. (2012) measured consumption rate as the amount of food consumed per gram of snail each day [$\text{mg g}^{-1} \text{d}^{-1}$]. This can be compared to the feeding rate in the model, which was calculated as the predicted food intake (i.e. the predicted consumed energy) per predicted gram of snail (wet weight) per day [$\text{J g}^{-1} \text{d}^{-1}$]. Because the actual

Table 3
Mathematical implementation of the DEB parameters under stress for the three investigated PMoAs (Jager and Zimmer 2012).

PMoA	Affected parameters	Stressed parameter
Decrease in assimilation	f	$f_s = f \times (1 - s)$
Increase in costs for growth	k_{Ms} , E_G	$k_{Ms} = k_M / (1 + s)$ $E_{Gs} = E_G \times (1 + s)$
Increase in costs for maintenance	k_M , k_J	$k_{Ms} = k_M \times (1 + s)$ $k_{Js} = k_J \times (1 + s)$

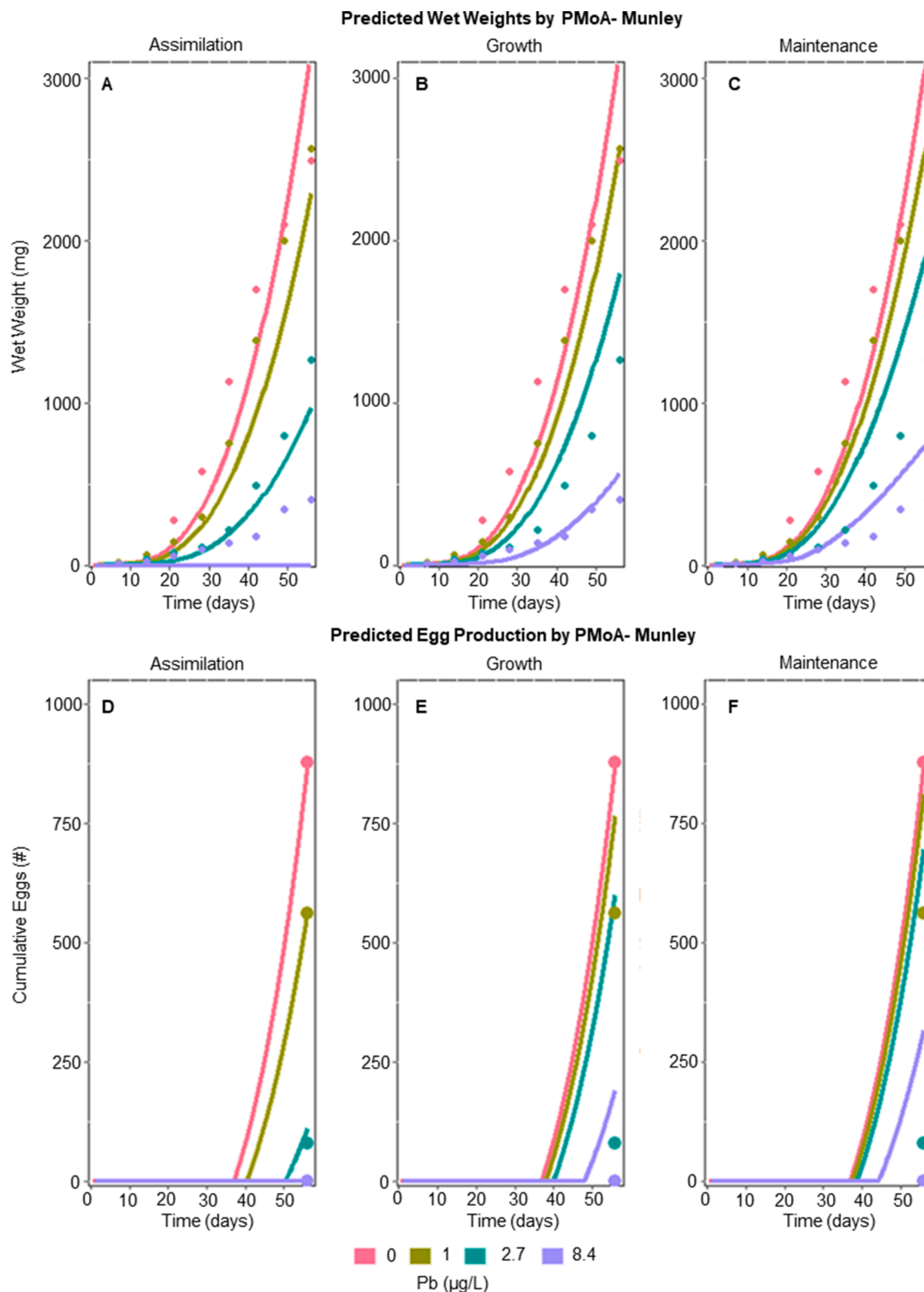


Fig. 2. Model predictions, as calibrated to data from [Munley et al. \(2013\)](#), for the assimilation, growth, and maintenance PMoAs. Data on experimentally observed snail weight and cumulative egg production (points) is plotted against the model predictions (lines of corresponding color). Time indicates time since hatching.

nutritional density of the food used in the experiment (J/mg food) was not known, we could not compare observations and prediction of absolute food intake. Therefore, we compared the observed consumption rate relative to control with the predicted assimilation rate relative to

control.

Of the three investigated PMoAs, only decreased assimilation was able to predict the observed decrease in consumption rate with increasing lead concentrations ([Fig. 4](#)). In simulations of decreased

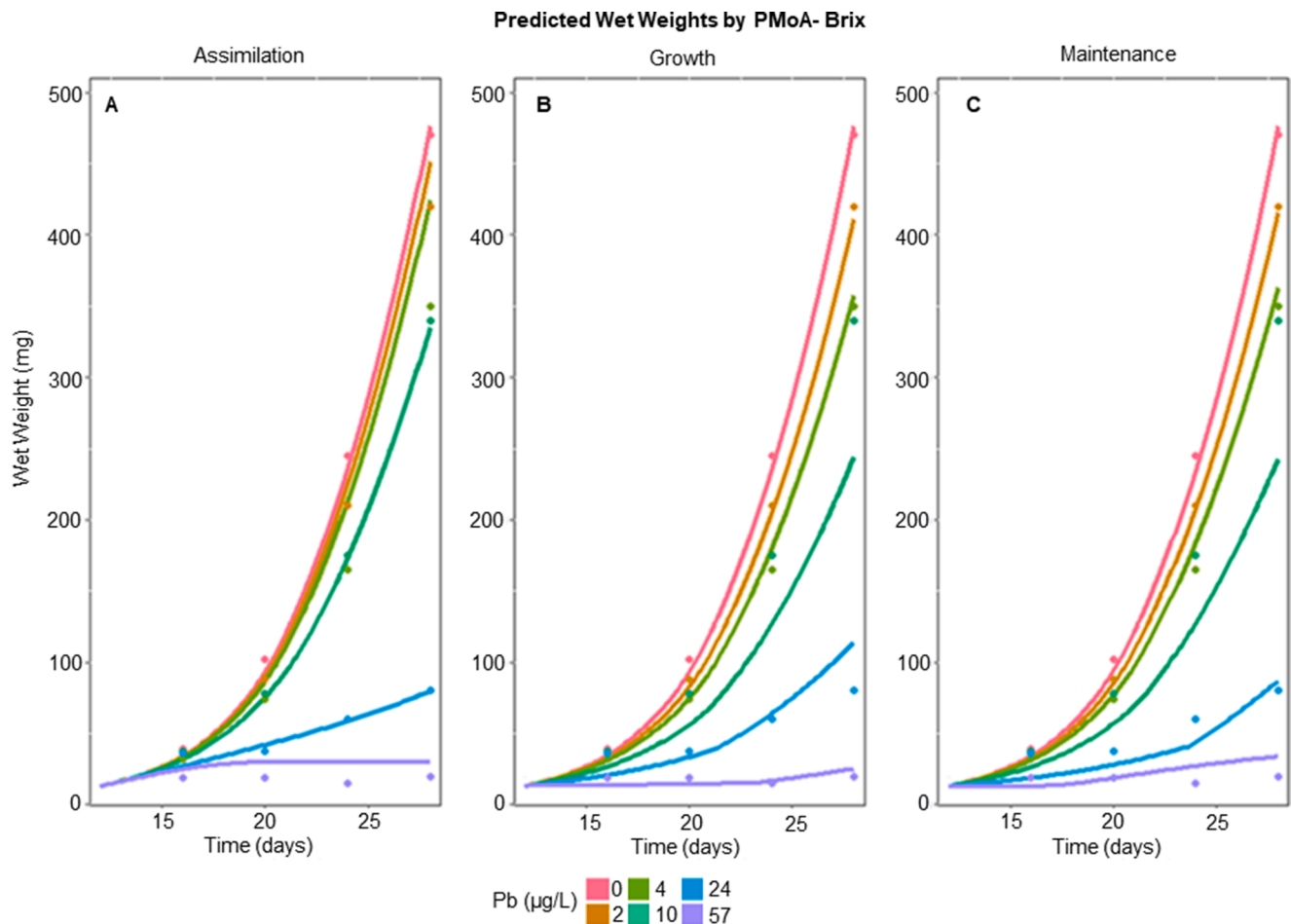


Fig. 3. Model predictions, as calibrated to data from Brix et al. (2012), for the assimilation, growth, and maintenance PMoAs. Data on experimentally observed snail weight (points) is plotted against the model predictions (lines of corresponding color). Time indicates time since hatching.

assimilation as the PMoA, the predicted decrease in assimilation rate, which scales with the surface-area of an organism, outpaced the predicted decrease in wet weight, which scales with volume. The opposite was true for simulations of both increased cost of growth and increased cost of maintenance, where increasing consumption rates (ingestion rate per gram snail) were predicted with increasing lead concentrations. Assuming decreased assimilation as the PMoA, predicted effects were slightly overestimated after 8 days, but correspondence of predictions with observations improved after 16 days of exposure.

3.3. Time to first egg production

In Munley et al. (2013), all experimental lead concentrations were found to have an impact on the time elapsed before snails began to produce eggs. Egg production was delayed in snails exposed to 1 or 2.7 µg/L of lead. For snails exposed to 2.7 µg/L of lead, the onset of reproduction was delayed by more than two weeks, relative to control snails. No egg production was observed in snails exposed to 8.4 µg/L of lead. For all experimental concentrations, the PMoA with decreased assimilation best predicted the effect of lead on the time to first egg production (Fig. 5). Simulations with increased growth costs or increased maintenance costs as the PMoA predicted egg production at 8.4 µg/L lead, which is contrary to experimental observations.

3.4. Recovery in egg production rate

Munley et al. (2013) noted a partial recovery pattern in the egg production of snails exposed to 1 µg/L of lead. Partial recovery was also

observed during the second two weeks of egg production in snails exposed to 2.7 µg/L lead. It should be noted that, while the term “recovery” is used here to describe model results, this is done for consistency of terminology with our focal study. This should not be interpreted as a modelled decrease in the chemical stress on an organism. While all three PMoAs predicted a trend toward recovery in reproductive end-points (Fig. 6), simulations of assimilation-stress most closely aligned with the observed recovery. As observed in Munley et al. (2013), simulations with decreased assimilation predicted that the egg production of snails at 2.7 µg/L trends toward recovery more slowly compared to 1 µg/L. This recovery trend only became apparent in the third week of egg production. The recovery pattern was not entirely captured by the model: at 1 µg/L lead, recovery stopped after 14 days in the predictions with decreased assimilation, while the observed recovery continued throughout the duration of the experiment. Nevertheless, the decreased assimilation gave substantially better predictions for this recovery than the other two PMoAs (Fig. 6).

3.5. Recovery in specific growth rate

Munley et al. (2013) found that snails exposed to 1 µg/L and 2.7 µg/L lead exhibited recovery in specific growth rate beginning from day 28. In fact, the specific growth rate of snails exposed to 1 µg/L lead was observed to ultimately exceed that of control snails. No such recovery in growth rate was observed in snails exposed to 8.4 µg/L lead. The pronounced growth rate recovery trend was not well captured by the DEB model as calibrated for any of the investigated PMoAs. This failure to predict growth recovery may be the result of the notable mismatch in

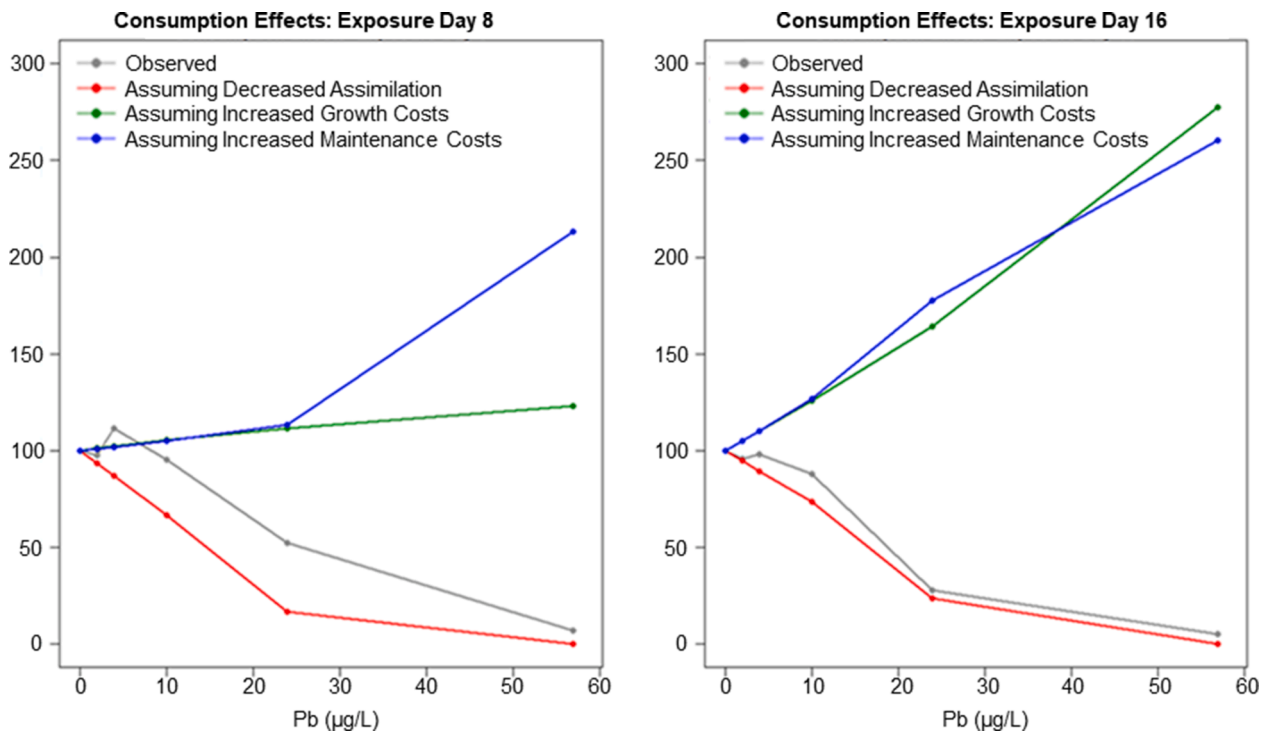


Fig. 4. Observed effect of lead exposure on snail food consumption rate [$\text{J g}^{-1} \text{d}^{-1}$] after 8 and 16 days of exposure (Brix et al., 2012), as compared to predicted effect on feeding rate assuming decreased assimilation, increased cost of growth, and increased cost of maintenance as the PMoA.

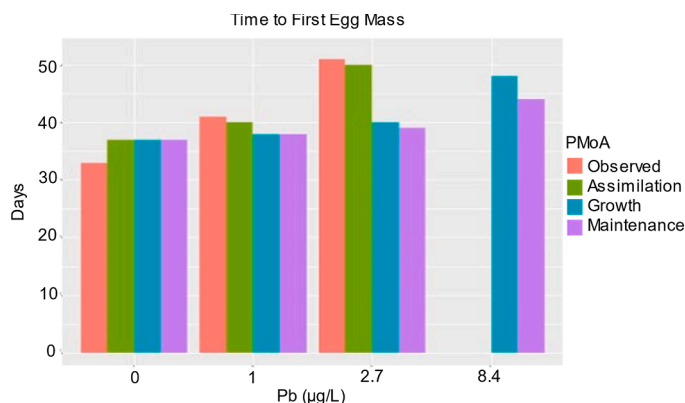


Fig. 5. Experimentally observed time to first egg production (red) at each lead concentration (Munley et al., 2013) compared to model predictions for assimilation, growth, and maintenance PMoAs.

growth pattern for the control calibration (Fig. 1).

4. Discussion

In the present study, we were able to calibrate DEB-TK parameters to individual level experimental data on lead toxicity to *L. stagnalis* from Munley et al. (2013) and Brix et al. (2012). In order to align model predictions of snail weight and reproduction with observed values for Munley et al. (2013), two model parameters, (1) the assimilation factor and (2) the maturity threshold at birth were re-calibrated. Once control observations could be reasonably predicted, we calibrated toxicity parameters for sublethal effects of lead assuming three distinct physiological modes of action: decreased assimilation, increased growth costs, and increased maintenance costs. Simulations assuming decreased assimilation resulted in the lowest error, supporting decreased assimilation as the PMoA of lead toxicity to *L. stagnalis*. Furthermore,

simulations of decreased assimilation best captured both the delay in the start of egg production with increasing lead concentrations and the observed recovery trend in egg production observed by Munley et al. (2013), which further supports this PMoA. However, it is important to note that the present analysis can only evaluate which of the three PMoAs investigated here best explains experimental observations. It certainly remains possible that the actual mechanism is different than those investigated or is, in fact, a combination of the PMoAs. Such combined mechanisms have been demonstrated as a possibility in previous studies (Ashauer and Jager 2018; Hansul et al., 2021), and may provide additional insight here as well.

Exposure to lead has previously been demonstrated to depress the feeding rate of exposed *L. stagnalis* snails relative to the snails' weight (Brix et al., 2012). After the model was calibrated to growth data from Brix et al. (2012), only the PMoA of decreased assimilation predicted a decrease in ingestion relative to weight. Both increased growth costs and increased maintenance costs resulted in an increase in assimilation relative to snail weight. The assimilated energy of a snail is the product of ingestion and assimilation efficiency. As such, the PMoA of decreased assimilation may capture patterns associated with a change in either. In this study, the predicted decrease in assimilation rate relative to snail weight was nearly identical to the observed decrease in consumption rate relative to snail weight by Day 16. This suggests that the inhibition of food ingestion itself, rather than assimilation efficiency, is the primary PMoA of lead in snails. If a decrease in assimilation efficiency was the primary driver of an overall decrease in assimilated energy, we would not expect the observed changes in ingestion to align with the predicted changes in assimilated energy in the model.

In addition to a reduction in net Ca^{2+} uptake, Brix et al. (2012) noted that a decrease in snail wet weights manifested at lower lead concentrations than the depression of ingestion. Consequently, it was concluded that the impact of lead on the ingestion rate of *L. stagnalis* could not be the exclusive driver of lead toxicity (Brix et al., 2012). The predictions of our mechanistic model yield different conclusions. Based on these, it is possible that decreased assimilation efficiency plays a greater role in determining the consequences of early exposure (or in

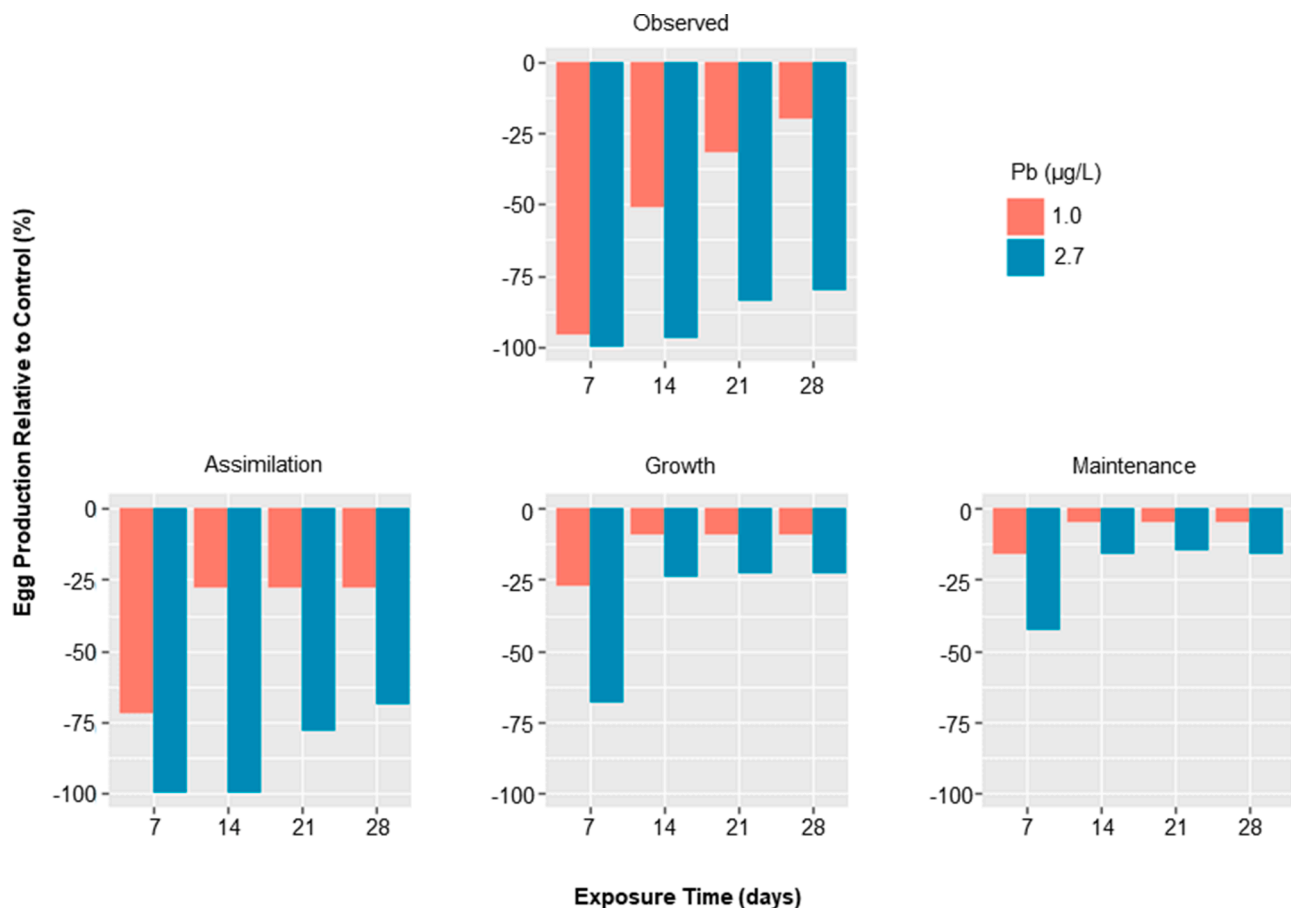


Fig. 6. Predicted egg production per week of adult *L. stagnalis* exposed to 1.0, and 2.7 µg/L of lead (relative to control), assuming each of the investigated PMoAs. Observed inhibition of reproduction is also plotted for reference (Munley et al., 2013).

younger snails). However, lowered ingestion alone is sufficient to predict effects on growth and consumption by day 16 of exposure. While Brix et al. (2012) investigated the mechanism of lead toxicity to *L. stagnalis* using a combined experimental and statistical approach, this study provides a complementary mechanistic modeling framework to relate the importance of ingestion versus assimilation efficiency to toxicity of lead. The nuance of the interplay between these two aspects of assimilation could be further investigated if assimilated energy relative to ingestion would be experimentally measured. A discrepancy remains between model predictions of decreased assimilation after eight days of exposure and the observed decrease in ingestion, which should remain a consideration in the weight of evidence. However, we currently propose decreased assimilation via decreased ingestion as the primary mechanism of lead toxicity to *L. stagnalis*.

Decreased assimilation (likely via decreased ingestion) as the primary driver of lead toxicity to *L. stagnalis* is supported by several independent lines of evidence. Assuming decreased assimilation as the PMoA, both the timing of first reproduction and later partial recovery in reproduction were captured by the model. Decreased assimilation was also the only investigated PMoA to accurately predict the observed decrease in ingestion rate. While the experimentally observed partial recovery in specific growth rate was not captured by any of the investigated PMoAs, this is unsurprising considering the mismatch in growth pattern between Munley et al. (2013) and the model (Supplemental Information). Snails observed in Munley et al. (2013) exhibited roughly linear growth in their bodyweights, which differs from the von Bertalanffy (with initial metabolic acceleration) growth pattern that is typically reported in the Add-my-pet database (and is therefore implemented in the model).

The present work provides additional insights into what the primary

PMoA of lead toxicity to *L. stagnalis* may be. These insights can be useful in future work considering lead toxicity to *L. stagnalis* at both lower and higher levels of biological organization. For instance, a variety of mechanisms, from molecular initiating events to behavioral mechanisms, can give rise to the different PMoA if their net effect is conceptualized as a change in the animal's energy budget. As such, the approach of this study cannot be used to identify a particular molecular initiating event. However, molecular initiating events which are linked with a decrease in assimilated energy may be considered more likely, given our results. Furthermore, identifying the mode of action of a given toxicant is beneficial in anticipating population level effects, which are of interest in environmental risk assessment. Indeed, a variety of studies have indicated that different PMoAs can result in the emergence of distinct compensation mechanisms (or a lack thereof) at the population level (Kooijman and Metz 1983; Martin et al., 2013a, Vlaeminck et al., 2022).

DEB (and mechanistic modeling more broadly) offers an opportunity to parse additional insights regarding physiological modes of action from existing datasets. By compiling data from two published studies (Brix et al., 2012; Munley et al., 2013), we were able to draw independent lines of evidentiary support for decreased assimilation as the primary mode of action of lead toxicity to *L. stagnalis*.

5. Conclusion

A partial re-parameterization of the DEB parameters for *L. stagnalis* resulted in reasonable growth and reproduction predictions. Using the reparametrized model, our predictions for the PMoA of decreased assimilation most closely aligned with experimental observations for lead effects on growth and reproductive output. Additionally, assuming

decreased assimilation, the model was able to predict some of the experimentally observed recovery patterns in reproductive endpoints. Finally, decreased assimilation as a consequence of decreased ingestion was the only PMoA able to predict the observed decrease in ingestion relative to snail weight. These results provide lines of support that, within the DEB framework, decreased assimilation via decreased ingestion is the most plausible physiological mode of action for lead toxicity in *L. stagnalis*.

CRedit authorship contribution statement

Kristi Weighman: Methodology, Software, Investigation, Visualization, Writing – original draft. **Karel Viaene:** Methodology, Conceptualization, Supervision, Writing – review & editing. **Josef Koch:** Methodology, Software, Investigation, Writing – review & editing. **Karel De Schampelaere:** Conceptualization, Supervision, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

Funding for this study was received from the International Lead Association and the Lead REACH Consortium. J. Koch was supported by the Research Foundation–Flanders through an SB PhD fellowship grant (project number: 1S06017N).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.aquatox.2023.106617](https://doi.org/10.1016/j.aquatox.2023.106617).

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